

Roll No.

TEC-504

**B. TECH. (ECE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

CONTROL SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Differentiate between open loop and closed loop control system. (CO1)

OR

- (b) Derive Force-Voltage analogy for mass spring dashpot mechanical system and parallel LCR circuit. (CO1)

2. (a) Derive an expression for peak time in case of unit step response for a second order unity feedback system. (CO1)

OR

- (b) Draw signal flow graph for the following set of simultaneous equations : (CO1)

$$X_2 = X_1 + 2X_2$$

$$X_3 = 2X_1 + 3X_2 + 4X_3$$

$$X_4 = 5X_2 + 6X_3$$

Also find transfer function using Mason's gain formula.

P. T. O.

(2)

3. (a) The closed loop transfer function of second order system is :

(CO2, CO3)

$$\frac{C(s)}{R(s)} = \frac{2}{s^2 + 3s + 2}$$

Determine the damping ratio, characteristic equation, poles and the natural frequency of oscillations. Also determine its unit step response.

OR

- (b) Derive the unit impulse response for a first order system. (CO2, CO3)

4. (a) Derive the expression for settling time for a unity feedback second order control system. (CO2)

OR

- (b) Discuss open loop control system and closed loop control system with suitable examples. (CO2)

5. (a) What are important features of a transfer function model. (CO1, CO2)

OR

- (b) Indicate rise time, peak time and settling time in an underdamped unit step response of a second order system. Also write formulae and the symbols used. (CO1, CO2)